
Los Sabrossos

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1. Math

1.1. Sieve of Eratosthenes

```
1  const int N = 1e7;
2  int sieve[N];
3  void make(){
4      for(int i = 0; i < N; i++) sieve[i] = 1;
5      sieve[0] = sieve[1] = 0;
6      for (int i = 2; i < N; i++)
7          {
8              if(sieve[i]){
9                  for (int j = i*i; j < N; j+=i)
10                     {
11                         sieve[j] = 0;
12                     }
13             }
14         }
15 }
```

1.2. Euler Phi

```
1  ll phi(ll n) {
2      ll ans = n;
3      for (ll p = 2; p * p <= n; p++) {
4          if (n % p == 0) {
5              ans -= ans / p;
6              while (n % p == 0){
7                  n /= p;
8              }
9          }
10     }
11     if (n > 1){
12         ans -= ans / n;
13     }
14     return ans;
15 }
```

1.3. Binary Exponentiation

```
1  long long pote(long long a, long long b, long long mod){
2      long long ans = 1ll;
3      while(b){
4          if(b&1) ans = (ans * a)%mod;
5          a = (a * a)%mod;
6          b>>=1;
7      }
8      return ans;
9  }
10 // a^-1 = a^mod-2 = pote(a,mod-2);
```

1.4. Chinese Rest Theorem

```
1  //si los modulos son muy grandes 1e18, cambiar por int128 las operaciones
2  ll extgcd(ll a, ll b, ll &x, ll &y) {
3      if (!b) {
4          x = 1;
5          y = 0;
6          return a;
7      }
8
9      ll x1, y1;
10     ll g = extgcd(b, a % b, x1, y1);
11
12     x = y1;
13     y = x1 - (a / b) * y1;
14
15     return g;
16 }
17
18 // x      a (mod m)
19 // x      b (mod n)
20 //si retorna {0, -1} -> no solution
21 pair<ll,ll> CRT(ll a, ll m, ll b, ll n) {
22     ll x, y;
```

```

23     ll g = extgcd(m, n, x, y);
24
25     if ((b - a) % g != 0)
26         return {0, -1};
27
28     ll n2 = n / g;
29
30     ll t = ((b - a) / g * x) % n2;
31     if (t < 0) t += n2;
32
33     ll mod = m * n2;
34     ll r = (a + m * t) % mod;
35     if (r < 0) r += mod;
36
37     return {r, mod};
38 }
39
40 // x      a[i] (mod m[i]) para todo i
41 //vector de valores y vector de respectivos modulos
42 pair<ll,ll> CRT(vector<ll> a, vector<ll> m) {
43     ll r = a[0];
44     ll mod = m[0];
45
46     for (int i = 1; i < a.size(); i++) {
47         pair<ll,ll> res = CRT(r, mod, a[i], m[i]);
48
49         if (res.second == -1)
50             return {0, -1};
51
52         r = res.first;
53         mod = res.second;
54     }
55
56     return {r, mod};
57 }
58
59 void solve(){
60     vector<ll>a(2);
61     vector<ll>m(2);
62     repl(i,0,2){
63         cin >> a[i] >> m[i];
64     }
65     pair<ll,ll> ans = CRT(a, m);
66     if(ans.second == -1){
67         cout << "-1\n";
68     }
69     else{
70         cout << ans.first << " " << ans.second << "\n";
71     }
72 }

```

1.5. Extended Euclides

```

1 //Encuentra
2 //valores para x e y
3 //A*x + B*y = gcd(A, B)
4 ll extgcd(ll a, ll b, ll &x, ll &y){
5     if(b == 0){
6         x = 1;
7         y = 0;
8         return a;
9     }
10
11     ll x1, y1;
12     ll g = extgcd(b, a % b, x1, y1);
13     x = y1;
14     y = x1 - (a / b) * y1;
15     return g;
16 }
17
18 void solve(){
19     ll A, B, C;
20     cin >> A >> B >> C;
21     ll x, y;
22     ll g = extgcd(A, B, x, y);
23     if(C % g != 0){
24         cout << "-1\n";
25         return;
26     }
27     ll k = -C / g;
28     x *= k;
29     y *= k;
30     cout << x << " " << y << "\n";
31 }

```

1.6. Fibonacci Log(n)

```
1 long long int fibo(int n) {
2     long long int a, b, x, y, tmp;
3     a = x = tmp = 0;
4     b = y = 1;
5     while (n != 0)
6         if (n % 2 == 0) {
7             tmp = a * a + b * b;
8             b = b * a + a * b + b * b;
9             a = tmp;
10            n = n / 2;
11        } else {
12            tmp = a * x + b * y;
13            y = b * x + a * y + b * y;
14            x = tmp;
15            n = n - 1;
16        }
17    // x = fibo(n-1)
18    return y; // y = fibo(n)
19 }
```

1.7. Least Common Multiple

```
1 ll lcm(ll a, ll b){
2     return (b/__gcd(a,b)) * a;
3 }
```

1.8. Modular Inverse

```
1 const long long N = 1e6 + 10;
2 const int MOD = 1e9+7;
3 long long fact[N];
4 long long invfact[N];
5 void init(){
6     fact[0] = 1;
7     for(long long i = 1; i < N; i++)
8         fact[i] = fact[i-1] * i % MOD;
9
10    invfact[N-1] = pote(fact[N-1], MOD-2);
11
12    for(long long i = N-2; i >= 0; i--)
13        invfact[i] = invfact[i+1] * (i+1) % MOD;
14 }
15
16 long long nCk(long long n, long long k){
17     if(k < 0 || k > n) return 0;
18     return fact[n] * invfact[k] % MOD * invfact[n-k] % MOD;
19 }
```

1.9. Pollard Rho Rabin Miller

```
1 map<ll, ll> fact;
2
3 ll pote(ll a, ll b, ll mod) {
4     ll ans = 1;
5     while (b) {
6         if (b & 1) ans = (__int128)ans * a % mod;
7         a = (__int128)a * a % mod;
8         b >>= 1;
9     }
10    return ans;
11 }
12
13 ll mcd(ll a, ll b) {
14     a = abs(a);
15     b = abs(b);
16     while (a != 0) {
17         ll r = b % a;
18         b = a;
19         a = r;
20     }
21    return b;
22 }
```

```
22 }
23
24 //Rabin Miller
25 bool primo(ll n) {
26     if (n < 2) return false;
27     if (n == 2) return true;
28     ll D = n - 1, S = 0;
29     while (D % 2 == 0) {
30         D /= 2;
31         S++;
32     }
33     static const ll STEPS = 1611;
34     for(ll pasos = 0; pasos < STEPS; pasos++){
35         const ll A = 1 + rand() % (n - 1);
36         ll M = pote(A, D, n);
37         if (M == 1 || M == (n - 1)) goto next;
38         for(ll k = 0; k < S-1; k++){
39             M = (__int128)M * M % n;
40             if (M == (n - 1)) goto next;
41         }
42         return false;
43     next:;
44     }
45     return true;
46 }
47
48 //Rho de pollard
49 ll factor(ll n) {
50     static ll A, B;
51     A = 1 + rand() % (n - 1);
52     B = 1 + rand() % (n - 1);
53     #define fun(x) (((__int128)(x) * ((x) + B)) % n + A)
54
55     ll x = 2, y = 2, d = 1;
56     while (d == 1 || d == -1) {
57         x = fun(x);
58         y = fun(fun(y));
59         d = mcd(x - y, n);
60     }
61     return abs(d);
62 }
63
64 void factorize(ll n){
65     assert(n > 0);
66     while (n > 1 && !primo(n)) {
67         ll fx;
68         do {
69             fx = factor(n);
70         } while (fx == n);
71
72         n /= fx;
73         factorize(fx);
74
75         for (auto &it : fact) {
76             while (n % it.first == 0) {
77                 n /= it.first;
78                 it.second++;
79             }
80         }
81     }
82     if (n > 1) fact[n]++;
83 }
```

2. Geometry

2.1. Convex Hull

```
1 // Devuelve sentido horario
2 vector<point> hull(vector<point> p)
3 {
4     int n = p.size();
5     vector<point> h;
6     sort(p.begin(), p.end());
7     for(int i = 0; i < n; i++){
8         while(h.size() >= 2 and p[i].left(h[h.size() - 2], h.back()) < 0){
9             h.pop_back();
10        }
11        h.push_back(p[i]);
12    }
13    h.pop_back();
14    int k = h.size();
```

```

15     for(int i = n-1; i >= 0; i--)
16     {
17         while(h.size() >= k + 2 and p[i].left(h[h.size() - 2], h.back()) < 0){
18             h.pop_back();
19         }
20         h.pb(p[i]);
21     }
22     h.pop_back();
23     return h;
24 }

```

2.2. Point

```

1  struct point
2  {
3      int x, y;
4      point() {}
5      point(int x, int y): x(x), y(y) {}
6      point operator -(point p) {return point(x - p.x, y - p.y);}
7      point operator +(point p) {return point(x + p.x, y + p.y);}
8      int sq() const {return x * x + y * y;}
9      double abs() const {return sqrt(sq());}
10     int operator ^(point p) {return x * p.y - y * p.x;}
11     int operator *(point p) {return x * p.x + y * p.y;}
12     point operator *(int a) {return point(x * a, y * a);}
13     bool operator <(const point& p) const {return x == p.x ? y < p.y : x < p.x;}
14     int left(point a, point b) {
15         return ((b - a) ^ (*this - a)); // 0, -, +
16     }
17 };
18 ostream& operator<<(ostream& os, const point& p) {
19     return os << "(" << p.x << ", " << p.y << ")\n";
20 }
21
22 void polarSort(vector<point>& v) {
23     sort(v.begin(), v.end(), [] (point a, point b) {
24         const point origin{0, 0};
25         bool ba = a < origin, bb = b < origin;
26         if (ba != bb) { return ba < bb; }
27         return (a^b) > 0;
28     });
29 }
30 }

```

2.3. Closest Pair of Points Lineal

```

1  struct pt {
2      ll x, y;
3      bool operator == (const pt & o) const {
4          return x == o.x and y == o.y;
5      }
6  };
7
8  struct CustomHashPoint {
9      size_t operator()(const pt & p) const {
10         static
11         const uint64_t C =
12             chrono::steady_clock::now().time_since_epoch().count();
13         uint64_t x = p.x;
14         uint64_t y = p.y;
15         x ^= x >> 30;
16         x *= 0xbf58476d1ce4e5b9;
17         x ^= x >> 27;
18         x *= 0x94d049bb133111eb;
19         x ^= x >> 31;
20         y ^= y >> 30;
21         y *= 0xbf58476d1ce4e5b9;
22         y ^= y >> 27;
23         y *= 0x94d049bb133111eb;
24         y ^= y >> 31;
25         return C ^ (x + (y << 1));
26     }
27 };
28
29 ll dist2(pt a, pt b) {
30     ll x = a.x - b.x;
31     ll y = a.y - b.y;
32     return x * x + y * y;
33 }
34
35 pair < int, int > closest_pair(vector < pt > p) {

```



```

36     int n = p.size();
37     unordered_map < pt, int, CustomHashPoint > mp;
38     // puntos repetidos
39     for (int i = 0; i < n; i++) {
40         if (mp.count(p[i]))
41             return {
42                 mp[p[i]],
43                 i
44             };
45         mp[p[i]] = i;
46     }
47     mt19937 rng(
48         chrono::steady_clock::now().time_since_epoch().count()
49     );
50     auto rnd = [&]() {
51         return uniform_int_distribution < int > (0, n - 1)(rng);
52     };
53     ll d2 = dist2(p[0], p[1]);
54     pair < int, int > ans = {
55         0,
56         1
57     };
58     auto upd = [&](int i, int j) {
59         ll nd = dist2(p[i], p[j]);
60         if (nd < d2) {
61             d2 = nd;
62             ans = {
63                 i,
64                 j
65             };
66         }
67     };
68     // conseguir una distancia inicial razonable
69     for (int it = 0; it < 20; it++) {
70         int i = rnd();
71         int j = rnd();
72         while (i == j) j = rnd();
73         upd(i, j);
74     }
75     ll d = sqrtl((long double) d2) + 1;
76     unordered_map < pt, vector < int >, CustomHashPoint > grid;
77     grid.reserve(2 * n);
78     for (int i = 0; i < n; i++) {
79         grid[
80             p[i].x / d,
81             p[i].y / d
82         ].push_back(i);
83     }
84     for (auto & [c, v]: grid) {
85         // misma celda
86         for (int i = 0; i < (int) v.size(); i++) {
87             for (int j = i + 1; j < (int) v.size(); j++) {
88                 upd(v[i], v[j]);
89             }
90         }
91         // celdas vecinas
92         for (int dx = -1; dx <= 1; dx++) {
93             for (int dy = -1; dy <= 1; dy++) {
94                 if (dx == 0 and dy == 0) continue;
95                 pt to = {
96                     c.x + dx,
97                     c.y + dy
98                 };
99                 auto it = grid.find(to);
100                 if (it == grid.end()) continue;
101                 for (int i: v) {
102                     for (int j: it->second) {
103                         upd(i, j);
104                     }
105                 }
106             }
107         }
108     }
109     return ans;
110 }

```

2.4. Closest Pair of Points Nlogn

```

1 struct pt {
2     ll x, y;
3     int id;
4 };
5
6 ll dist2(pt a, pt b) {

```

```

7     ll x = a.x - b.x;
8     ll y = a.y - b.y;
9     return x * x + y * y;
10 }
11
12 pair < int, int > closest_pair(vector < pt > p) {
13     int n = p.size();
14     sort(p.begin(), p.end(), [](pt a, pt b) {
15         if (a.x != b.x) return a.x < b.x;
16         return a.y < b.y;
17     });
18     set < pair < ll, int >> s; // {y, id}
19     ll d2 = dist2(p[0], p[1]);
20     pair < int, int > ans = {
21         p[0].id,
22         p[1].id
23     };
24     int l = 0;
25     for (int i = 0; i < n; i++) {
26         ll d = sqrtl((long double) d2) + 1;
27         // sacar puntos que ya est n demasiado lejos en x
28         while (l < i and p[i].x - p[l].x >= d) {
29             s.erase({
30                 p[l].y,
31                 p[l].id
32             });
33             l++;
34         }
35         // buscar puntos con y cerca
36         auto it = s.lower_bound({
37             p[i].y - d,
38             -1
39         });
40         while (it != s.end() and it -> first <= p[i].y + d) {
41             int j = it -> second;
42             ll nd = dist2(p[i], p[j]);
43             if (nd < d2) {
44                 d2 = nd;
45                 ans = {
46                     p[i].id,
47                     p[j].id
48                 };
49                 d = sqrtl((long double) d2) + 1;
50             }
51             it++;
52         }
53         s.insert({
54             p[i].y,
55             p[i].id
56         });
57     }
58     return ans;
59 }

```

3. Strings

3.1. KMP

```

1 vector<int> kmp(string s){
2     int n = s.size();
3     vector<int> pi(n);
4     for(int i = 1; i < n; i++){
5         int j = pi[i-1];
6         while(j > 0 and s[i] != s[j]){
7             j = pi[j-1];
8         }
9         if(s[i] == s[j]){
10             j++;
11         }
12         pi[i] = j;
13     }
14     return pi;
15 }
16 //Pattern Matching
17 vector<int> ocurrencia(string texto, string patron) {
18     vector<int> P = kmp(patron);
19     int k = 0;
20     vector<int> M;
21     repl(i,0,texto.size()) {
22         while (k > 0 and patron[k] != texto[i]) k = P[k - 1];
23         if (patron[k] == texto[i]) k++;

```

```

24         if (k == patron.size()) {
25             M.pb(i - k + 1);
26             k = P[k - 1];
27         }
28     }
29     return M;
30 }
31
32 //Automata
33 void genAut(string &s) {
34     int n = s.size();
35
36     vector<vector<int>>> aut(n+1, vector<int>(26));
37     vector<int> pi = kmp(s);
38
39     for(int i = 0 ; i < n; i++){
40         aut[i][s[i]-'a'] = i+1;
41     }
42
43     for(int i = 0; i < n+1; i++){
44         for(int c = 0; c < 26; c++){
45             if (aut[i][c]) continue;
46             if (i < n and s[i] == 'a' + c) aut[i][c] = i+1;
47             else if (i > 0) aut[i][c] = aut[pi[i-1]][c];
48             else aut[i][c] = 0;
49         }
50     }
51 }

```

3.2. Z Function

```

1  vector<int> z_function(string s){
2      int n = s.size();
3      vector<int> z(n,0);
4      int l = 0;
5      int r = 0;
6      repl(i,l,n){
7          if(i < r){
8              z[i] = min(r - i, z[i - l]);
9          }
10         while(i + z[i] < n and s[z[i]] == s[i + z[i]]){
11             z[i]++;
12         }
13         if(i + z[i] > r){
14             l = i;
15             r = i + z[i];
16         }
17     }
18     return z;
19 }

```

3.3. Trie

```

1  const int E = 26;
2
3  struct AC {
4      int nodes;
5      vector<array<int, E>>go;
6      vector<bool>terminal;
7
8      AC(){
9          add_node();
10         nodes = 1;
11     }
12
13     void add_node(){
14         terminal.emplace_back();
15         go.emplace_back();
16     }
17
18     void insert(string &s){
19         int pos = 0;
20         for(char ch : s){
21             int c = ch - '0';
22             if(go[pos][c] == 0){
23                 go[pos][c] = nodes++;
24                 add_node();
25             }
26             pos = go[pos][c];
27         }
28         terminal[pos] = true;

```

```

29     }
30
31 };

```

3.4. Suffix Array

```

1  vector<int> suffix_array(string s) {
2      s += char(0);
3      const int n = s.size();
4      vector<int> a(n);
5      iota(a.begin(), a.end(), 0);
6      sort(a.begin(), a.end(), [&](int i, int j){
7          return s[i] < s[j];
8      });
9      vector<int> mapping(n);
10     mapping[a[0]] = 0;
11     for(int i = 1; i < n; i++){
12         mapping[a[i]] = mapping[a[i-1]] + (s[a[i-1]] != s[a[i]]);
13     }
14     int len = 1;
15     vector<int> sbs(n);
16     vector<int> head(n);
17     vector<int> new_mapping(n);
18     while(len < n){
19         for(int i = 0; i < n; i++) sbs[i] = (a[i] + n - len) % n;
20         for(int i = n-1; i >= 0; i--) head[mapping[a[i]]] = i;
21         for(int i = 0; i < n; i++){
22             int x = sbs[i];
23             a[head[mapping[x]]++] = x;
24         }
25         new_mapping[a[0]] = 0;
26         for(int i = 1; i < n; i++){
27             if(mapping[a[i-1]] != mapping[a[i]]){
28                 new_mapping[a[i]] = new_mapping[a[i-1]] + 1;
29             }
30             else{
31                 int pre = mapping[(a[i-1] + len) % n];
32                 int cur = mapping[(a[i] + len) % n];
33                 new_mapping[a[i]] = new_mapping[a[i-1]] + (pre != cur);
34             }
35         }
36         len <<= 1;
37         swap(mapping, new_mapping);
38     }
39     return vector<int>(a.begin()+1, a.end()); //ignorar a[0] que es el centinela
40 }
41
42 vector<int> lcp_array(vector<int> &a, string s){
43     const int n = s.size();
44     vector<int> rank(n);
45     for(int i = 0; i < n; i++) rank[a[i]] = i;
46     int k = 0;
47     vector<int> lcp(n);
48     for(int i = 0; i < n; i++){
49         if(rank[i] == n-1){
50             k = 0;
51             continue;
52         }
53         int j = a[rank[i]+1];
54         while(i+k < n and j+k < n and s[i+k] == s[j+k]) k++;
55         lcp[rank[i]] = k;
56         if(k) k--;
57     }
58     lcp.pop_back();
59     return lcp;
60 }
61
62 void solve(){
63     //Numero de distintos substr en un string n*(n+1)/2 - sumatoria del LCP
64     //verificar si un string es substring de otro con BS en SA
65     //encontrar el longest common substring en 2 strings (ESTA IMPLEMENTACION)
66     string s, t;
67     cin >> s >> t;
68     if(s.size() > t.size()){
69         swap(s, t);
70     }
71     int n = s.size();
72     int m = t.size();
73     string st = s + '#' + t;
74     vector<int> sa = suffix_array(st);
75     vector<int> lcp = lcp_array(sa, st);
76     // print(sa);
77     // print(lcp);
78     vector<int> tr(n+m);

```

```

79     repl(i,0,n+m){
80         if(sa[i] >=n and sa[i+1]>=n){
81             tr[i] = 0;
82             continue;
83         }
84         if(sa[i] < n and sa[i+1] < n){
85             tr[i] = 0;
86             continue;
87         }
88         int mn = min(sa[i], sa[i+1]);
89         int tam1 = mn + lcp[i];
90         // dbg(tam1);
91         tr[i] = lcp[i];
92         if(tam1 > n){
93             dbg(tam1);
94             tr[i] = lcp[i] - (tam1-n);
95         }
96     }
97     // print(tr);
98     int mx = *max_element(tr.begin(), tr.end());
99     repl(i,0,n+m){
100         // dbg(tr[i]);
101         if(tr[i] == mx){
102             string ans = st.substr(sa[i], mx);
103             cout << ans << '\n';
104             return;
105         }
106     }
107 }
108 //Sabrossus

```

3.5. Aho Corasick

```

1 //AhoCorasick para saber si un string s existe en un texto T
2 const int E = 26;
3 const int N = 1e6+10;
4 int cur;
5 int nodoTerminal[N]; //se guarda el nodo terminal del string i
6 set<int>st; //se guarda que nodos son terminales que aparecen en el texto T
7
8 struct AC {
9     int nodes;
10    vector<int>suf, super;
11    vector<array<int, E>>go;
12    vector<bool>terminal;
13    AC(){
14        add_node();
15        nodes = 1;
16    }
17    void add_node(){
18        terminal.emplace_back();
19        go.emplace_back();
20        super.emplace_back();
21        suf.emplace_back();
22    }
23
24    void insert(string &s){
25        int pos = 0;
26        for(char ch : s){
27            int c = ch - 'a';
28            if(go[pos][c] == 0){
29                go[pos][c] = nodes++;
30                add_node();
31            }
32            pos = go[pos][c];
33        }
34        terminal[pos] = true;
35        nodoTerminal[cur] = pos;
36    }
37
38    void build(){
39        queue<int>Q;
40        Q.emplace(0);
41        while(!Q.empty()){
42            int u = Q.front();
43            Q.pop();
44            super[u] = (suf[u] == 0 or terminal[suf[u]] ? suf[u] : super[suf[u]]);
45            for(int c = 0; c < E; c++){
46                if(go[u][c]){
47                    int v = go[u][c];
48                    Q.emplace(v);
49                    suf[v] = u == 0 ? 0 : go[suf[u]][c];
50                }

```

```

51         else{
52             go[u][c] = u == 0 ? 0 : go[suf[u]][c];
53         }
54     }
55 }
56 }
57 };
58
59 void mark(int u, AC &ac){
60     if(u == 0) return;
61     mark(ac.super[u], ac);
62     if(ac.terminal[u]){
63         st.insert(u);
64         ac.terminal[u] = false;
65     }
66     ac.super[u] = 0;
67 }
68
69 void solve(){
70     int n;
71     cin >> n;
72     AC ac;
73     repl(i,0,n){
74         string s;
75         cin >> s;
76         cur = i;
77         ac.insert(s);
78     }
79     string t; //texto grande
80     cin >> t;
81     ac.build();
82     int p = 0;
83     for(auto x : t){
84         int c = x - 'a';
85         p = ac.go[p][c];
86         mark(p, ac);
87     }
88     repl(i,0,n){
89         if(st.count(nodoTerminal[i])){
90             cout << "YES\n";
91         }
92         else{
93             cout << "NO\n";
94         }
95     }
96 }
97
98 //-----
99
100 //AhoCorasick para Frecuencias
101 const int E = 26;
102 const int N = 1e6+10;
103 int nodoTerminal[N];
104 int cur;
105 struct AC {
106     int nodes;
107     vector<int>suf, freq, by_level;
108     vector<bool>terminal;
109     vector<array<int,E>>go;
110
111     AC(){
112         add_node();
113         nodes = 1;
114     }
115     void add_node(){
116         terminal.emplace_back();
117         suf.emplace_back();
118         go.emplace_back();
119         freq.emplace_back();
120     }
121
122     void insert(string &s){
123         int pos = 0;
124         for(char ch:s){
125             int c = ch - 'a';
126             if(go[pos][c] == 0){
127                 go[pos][c] = nodes++;
128                 add_node();
129             }
130             pos = go[pos][c];
131         }
132         terminal[pos] = true;
133         nodoTerminal[cur] = pos;
134     }
135
136     void build(){

```

```

137     queue<int>Q;
138     Q.emplace(0);
139     while(!Q.empty()){
140         int u = Q.front();
141         Q.pop();
142         by_level.emplace_back(u);
143         for(int c = 0; c < E; c++){
144             if(go[u][c]){
145                 int v = go[u][c];
146                 Q.emplace(v);
147                 suf[v] = u == 0 ? 0 : go[suf[u]][c];
148             }
149             else{
150                 go[u][c] = u == 0 ? 0 : go[suf[u]][c];
151             }
152         }
153     }
154 }
155 };
156
157 void solve(){
158     int n;
159     cin >> n;
160     AC ac;
161     repl(i,0,n){
162         string s;
163         cin >> s;
164         cur = i;
165         ac.insert(s);
166     }
167     string t;
168     cin >> t;
169     ac.build();
170     int p = 0;
171     for(char ch:t){
172         int c = ch-'a';
173         p = ac.go[p][c];
174         ac.freq[p]++;
175     }
176     for(int i = (int)ac.by_level.size()-1; i > 0; i--){
177         int x = ac.by_level[i];
178         ac.freq[ac.suf[x]] += ac.freq[x];
179     }
180     repl(i,0,n){
181         int termi = nodoTerminal[i];
182         cout << ac.freq[termi] << '\n';
183     }
184 }

```

3.6. Booths Algorithm

```

1 //Algoritmo para menor rotacion lexicografica de un string en O(N)
2
3 int menorRotacion(string &s){
4     int n = s.size();
5     vector<int>f(2*n, -1);
6     int k = 0; //indice mejor rotacion
7     for(int j = 1; j < 2*n;j++){
8         int i = f[j-k-1];
9         //el operador %n simula la concatenacion de la cadena s+s
10        while(i!=-1 and s[j%n] != s[(k+i+1)%n]){
11            if(s[j%n] < s[(k+i+1)%n]){
12                k = j - i - 1;
13            }
14            i = f[i];
15        }
16        if(i == -1 and s[j%n] != s[(k+i+1)%n]){
17            if(s[j % n] < s[(k+i+1)%n]){
18                k = j;
19            }
20            f[j - k] = -1;
21        }
22        else{
23            f[j - k] = i + 1;
24        }
25    }
26    return k;
27 }
28 void solve(){
29     string s;
30     cin >> s;
31     ll n = s.size();
32     ll ans = menorRotacion(s);
33     repl(i, ans, ans+n){

```

```

34         cout << s[i%n];
35     }
36 }

```

3.7. Hashing 2D

```

1  /*
2   2D Hashing: buscar todas las apariciones de una matriz
3   r x c dentro de una matriz n x m.
4   */
5
6   const ll BASE = 911382323LL;
7
8   void solve() {
9       int r, c;
10      cin >> r >> c;
11
12      vector<string> pattern(r);
13      for (int i = 0; i < r; i++)
14          cin >> pattern[i];
15
16      // Potencias para el rolling vertical.
17      vector<ll> power(r + 1, 1);
18      for (int i = 1; i <= r; i++)
19          power[i] = power[i - 1] * BASE;
20
21      // Hash de la matriz patr n.
22      ll patternHash = 0;
23
24      for (int i = 0; i < r; i++) {
25          Hash h(pattern[i]);
26          ll rowHash = h.get(0, c);
27
28          patternHash += rowHash * power[r - i - 1];
29      }
30
31      int n, m;
32      cin >> n >> m;
33
34      vector<string> grid(n);
35      for (int i = 0; i < n; i++)
36          cin >> grid[i];
37
38      // rowHash[i][j] = hash de grid[i][j ... j+c-1].
39      vector<vector<ll>> rowHash(
40          n, vector<ll>(m - c + 1)
41      );
42
43      for (int i = 0; i < n; i++) {
44          Hash h(grid[i]);
45
46          for (int j = 0; j <= m - c; j++)
47              rowHash[i][j] = h.get(j, j + c);
48      }
49
50      // matrixHash[i][j] = hash de la submatriz r x c desde (i, j).
51      vector<vector<ll>> matrixHash(
52          n - r + 1,
53          vector<ll>(m - c + 1)
54      );
55
56      for (int j = 0; j <= m - c; j++) {
57          ll cur = 0;
58
59          // Primera ventana vertical.
60          for (int i = 0; i < r; i++)
61              cur += rowHash[i][j] * power[r - i - 1];
62
63          matrixHash[0][j] = cur;
64
65          // Rolling vertical.
66          for (int i = 1; i <= n - r; i++) {
67              cur -= rowHash[i - 1][j] * power[r - 1];
68              cur = cur * BASE + rowHash[i + r - 1][j];
69
70              matrixHash[i][j] = cur;
71          }
72      }
73
74      // 2D difference array para marcar las apariciones.
75      vector<vector<int>> diff(
76          n + 1, vector<int>(m + 1, 0)
77      );
78

```



```

79     for (int i = 0; i <= n - r; i++) {
80         for (int j = 0; j <= m - c; j++) {
81             if (matrixHash[i][j] == patternHash) {
82                 diff[i][j]++;
83                 diff[i][j + c]--;
84                 diff[i + r][j]--;
85                 diff[i + r][j + c]++;
86             }
87         }
88     }
89 }
90
91 // Prefijos 2D.
92 for (int i = 0; i < n; i++)
93     for (int j = 1; j < m; j++)
94         diff[i][j] += diff[i][j - 1];
95
96 for (int j = 0; j < m; j++)
97     for (int i = 1; i < n; i++)
98         diff[i][j] += diff[i - 1][j];
99
100 // Imprimir celdas pertenecientes a alguna aparici n.
101 for (int i = 0; i < n; i++) {
102     for (int j = 0; j < m; j++) {
103         cout << (diff[i][j] > 0 ? grid[i][j] : '.');
104     }
105     cout << '\n';
106 }
107 }

```

3.8. Hashing

```

1  struct Hash {
2      long long P=1777771,MOD[2],PI[2];
3      vector<long long> h[2],pi[2];
4
5      // SOLO PARA ROLLING / JOIN:
6      vector<long long> pw[2];
7
8      Hash(string& s){
9          MOD[0]=999727999;
10         MOD[1]=107077777;
11         PI[0]=325255434;
12         PI[1]=10018302;
13
14         for(int k = 0; k < 2; k++) {
15             h[k].resize(s.size()+1);
16             pi[k].resize(s.size()+1);
17
18             // SOLO PARA ROLLING / JOIN:
19             pw[k].resize(s.size()+1);
20         }
21
22         for(int k = 0; k < 2; k++){
23             h[k][0] = 0;
24             pi[k][0] = 1;
25
26             // SOLO PARA ROLLING / JOIN:
27             pw[k][0] = 1;
28
29             long long p = 1;
30
31             for(int i = 1; i < s.size()+1; i++){
32                 h[k][i] = (h[k][i-1]+p*s[i-1])%MOD[k];
33                 pi[k][i] = (1LL*pi[k][i-1]*PI[k])%MOD[k];
34
35                 // SOLO PARA ROLLING / JOIN:
36                 pw[k][i] = (1LL*pw[k][i-1]*P)%MOD[k];
37
38                 p = (p*P)%MOD[k];
39             }
40         }
41     }
42
43     long long get(int s, int e){
44         long long h0 = (h[0][e]-h[0][s]+MOD[0])%MOD[0];
45         h0 = (1LL*h0*pi[0][s])%MOD[0];
46
47         long long h1 = (h[1][e]-h[1][s]+MOD[1])%MOD[1];
48         h1 = (1LL*h1*pi[1][s])%MOD[1];
49
50         return (h0<<32)|h1;
51     }
52 }

```

```
53 // SOLO PARA ROLLING / JOIN:
54 long long join(long long a, long long b, int lenA){
55     long long a0 = a>>32;
56     long long a1 = a&0xffffffffLL;
57
58     long long b0 = b>>32;
59     long long b1 = b&0xffffffffLL;
60
61     a0 = (a0 + b0*pw[0][lenA])%MOD[0];
62     a1 = (a1 + b1*pw[1][lenA])%MOD[1];
63
64     return (a0<<32)|a1;
65 }
66 };
```

3.9. Manacher

```
1  const int N = 1e5 + 10;
2  int odd[N]; //odd[i] = max odd palindrome centered on i
3  int even[N]; //even[i] = max even palindrome centered on i
4  void manacher(string & s) {
5      int l = 0, r = -1, n = s.size();
6      for (int i = 0; i < n; i++) {
7          int k = i > r ? 1 : min(odd[l + r - i], r - i);
8          while (i + k < n and i - k >= 0 and s[i + k] == s[i - k]) k++;
9          odd[i] = k--;
10         if (i + k > r) l = i - k, r = i + k;
11     }
12     l = 0;
13     r = -1;
14     for (int i = 0; i < n; i++) {
15         int k = i > r ? 0 : min(even[l + r - i + 1], r - i + 1);
16         k++;
17         while (i + k <= n and i - k >= 0 and s[i + k - 1] == s[i - k]) k++;
18         even[i] = --k;
19         if (i + k - 1 > r) l = i - k, r = i + k - 1;
20     }
21 }
```

4. Bit Manipulation

4.1. Binary Representation

```
1  string tobit(long long x) {
2      string a(64, '0');
3      for (int i = 63; i >= 0; --i) {
4          if (x & 1) a[i] = '1';
5          x >>= 1;
6      }
7      return a;
8  }
```

4.2. Bit Mask

```
1  for(int mask = 0; mask < (1 << n); mask++) { //(1 << n) = 0(2^n)
2      for(int i = 0; i < n; i++) { // 0(n)
3          //verificamos si el i-esimo bit esta encendido
4          if(mask & (1 << i)) {
5              cout << v[i] << " ";
6          }
7      }
8  }
```

4.3. Bits Prendidos En Un Numero X En Una Posicion Pos

```
1  //retorna el numero de bits prendidos
2  //en una posicion pos desde 0 hasta el numero num
3  //tambien se puede usar en rangos asi:
4  //prendidos(R, 1) - prendidos(L-1, 1);
5  ll prendidos(ll num, ll pos){
6      if(num <= 0){
7          return 0;
```

```
8     }
9     ll pote = (1ll << pos);
10    ll blocks = (num + 1) / pote;
11    ll ones = blocks / 2;
12    ones *= pote;
13    if(blocks & 1){
14        ones += (num + 1) % pote;
15    }
16    return ones;
17 }
18 //Sabrosson
```

4.4. Turn Off a Bit at Position X

```
1 void solve(){
2     //mapa de caracteres en windows para ~
3     int a, x;
4     cin >> a >> x; // a es el numero, x es el numero de bit
5     //los bits se enumeran de dercha a izquierda iniciando de 0
6     /*
7     Ej: n = 8
8     Binario:      1 0 0 0
9     Posicion:     3 2 1 0
10    */
11    a = a & ~(1 << x); //apagamos el bit x de a
12    cout << a << "\n";
13 }
```

4.5. Xor Basis

```
1 ll base[LOG];
2 ll tam = 0;
3 void add(ll msk){
4     for(int i = LOG-1; i>= 0; i--){
5         if(!(msk&(1LL<<i)))continue;
6         if(!base[i]){
7             base[i]=msk;
8             tam++;
9             return;
10        }
11        msk^=base[i];
12    }
13 }
```

5. Data Structures

5.1. Segment Tree

```
1 const int N = 1e5;
2
3 struct info {
4     int num;
5 };
6
7 info ST[4 * N];
8
9 info ope(info u, info v) {
10     return {u.num + v.num};
11 }
12
13 void build(const vector<int> &v, int in = 1, int L = 0, int R = n - 1) {
14     if (L == R) {
15         ST[in] = {v[L]}; // Asignamos el valor desde el vector
16     } else {
17         int mid = (L + R) >> 1;
18         build(v, 2 * in, L, mid);
19         build(v, (2 * in) + 1, mid + 1, R);
20         ST[in] = ope(ST[2 * in], ST[(2 * in) + 1]);
21     }
22 }
23
24 void update(int pos, int val, int in = 1, int L = 0, int R = n - 1) {
25     if (L == R) {
26         ST[in].num = val;
27     } else {
```

```

28     int mid = (L + R) >> 1;
29     if (pos <= mid) {
30         update(pos, val, 2 * in, L, mid);
31     } else {
32         update(pos, val, (2 * in) + 1, mid + 1, R);
33     }
34     ST[in] = ope(ST[2 * in], ST[(2 * in) + 1]);
35 }
36 }
37
38 info get(int l, int r, int in = 1, int L = 0, int R = n - 1) {
39     if (r < L or R < l) return {0}; // Elemento neutro
40     if (l <= L and R <= r) return ST[in];
41
42     int mid = (L + R) >> 1;
43     info left = get(l, r, 2 * in, L, mid);
44     info right = get(l, r, (2 * in) + 1, mid + 1, R);
45     return ope(left, right);
46 }
47
48 // Llamadas simples y limpias:
49 // build(v); // Construye con el vector
50 // update(2, 10); // Cambia el valor en pos 2 a 10
51 // info ans = get(1, 3); // Obtiene la respuesta en el rango [1, 3]
52 // v = {1 2 3 4 5}
53 // get(1, 0, n-1, 1, 3) = 2 + 3 + 4 = 9

```

5.2. Segment Tree Iterativo

```

1  struct info
2  {
3      int num;
4  };
5
6  info ope(info u, info v){
7      return {min(u.num, v.num)};
8  }
9
10 int sz;
11 vector<info> ST;
12
13 void build(int n) {
14     sz = 1;
15     while (sz < n) sz *= 2;
16     ST.assign(2 * sz, {1000000000});
17 }
18
19 void update(int pos, int val) {
20     pos += sz;
21     ST[pos] = ope(ST[pos], {val});
22     for (pos /= 2; pos > 0; pos /= 2) {
23         ST[pos] = ope(ST[2 * pos], ST[2 * pos + 1]);
24     }
25 }
26
27 info get(int l, int r) {
28     info res = {1000000000};
29     for (l += sz, r += sz + 1; l < r; l /= 2, r /= 2) {
30         if (l & 1) res = ope(res, ST[l++]);
31         if (r & 1) res = ope(res, ST[--r]);
32     }
33     return res;
34 }
35 //Las operaciones y consultas son iguales al recursivo, 0-index

```

5.3. Segment Tree Lazy Propagation

```

1  const int N = 1e5;
2
3  struct info {
4      long long num;
5  };
6
7  info ST[4 * N];
8  long long lazy[4 * N];
9
10 info ope(info u, info v) {
11     return {u.num + v.num};
12 }
13

```

```

14 void build(const vector<int> &v, int in = 1, int L = 0, int R = n - 1) {
15     if (L == R) {
16         ST[in] = {v[L]};
17     } else {
18         int mid = (L + R) >> 1;
19
20         build(v, 2 * in, L, mid);
21         build(v, 2 * in + 1, mid + 1, R);
22
23         ST[in] = ope(ST[2 * in], ST[2 * in + 1]);
24     }
25 }
26
27 void push(int in, int L, int R) {
28     if (lazy[in] == 0)
29         return;
30
31     ST[in].num += lazy[in] * (R - L + 1);
32
33     if (L != R) {
34         lazy[2 * in] += lazy[in];
35         lazy[2 * in + 1] += lazy[in];
36     }
37
38     lazy[in] = 0;
39 }
40
41 void update(int l, int r, long long val,
42             int in = 1, int L = 0, int R = n - 1) {
43
44     push(in, L, R);
45
46     if (r < L or R < l)
47         return;
48
49     if (l <= L and R <= r) {
50         lazy[in] += val;
51         push(in, L, R);
52         return;
53     }
54
55     int mid = (L + R) >> 1;
56
57     update(l, r, val, 2 * in, L, mid);
58     update(l, r, val, 2 * in + 1, mid + 1, R);
59
60     ST[in] = ope(ST[2 * in], ST[2 * in + 1]);
61 }
62
63 info get(int l, int r,
64          int in = 1, int L = 0, int R = n - 1) {
65
66     push(in, L, R);
67
68     if (r < L or R < l)
69         return {0};
70
71     if (l <= L and R <= r)
72         return ST[in];
73
74     int mid = (L + R) >> 1;
75
76     info left = get(l, r, 2 * in, L, mid);
77     info right = get(l, r, 2 * in + 1, mid + 1, R);
78
79     return ope(left, right);
80 }

```

5.4. Bit 1D

```

1 struct BIT{
2     int n;
3     vector<ll> bit;
4     BIT(int _n){
5         n = _n;
6         bit.assign(n + 1, 0);
7     }
8     void add(int x, ll val){
9         for (int i = x; i <= n; i += (i&(-i))){
10             bit[i] += val;
11         }
12     }
13 }
14 ll sum(int x){

```

```

15     ll ans = 0ll;
16     for (int i = x; i > 0; i -= (i&(-i))){
17         ans += bit[i];
18     }
19     return ans;
20 }
21 ll query(int l, int r){
22     return sum(r) - sum(l - 1);
23 }
24 };
25 // Para actualizar en rango y consultas puntuales
26 // bit.add(l, val); bit.add(r+1, val);
27 // bit.query(l, pos)

```

5.5. Bit 2D

```

1 struct BIT{
2     int n, m;
3     vector<vector<ll>> bit;
4
5     BIT(int _n){
6         n = _n;
7         m = _n;
8         bit.assign(n+1, vector<ll>(n+1, 0));
9     }
10    void add(int x, int y, ll val){
11        for (int i = x; i <= n; i += (i&(-i))){
12            for (int j = y; j <= m; j += (j&(-j))){
13                bit[i][j] += val;
14            }
15        }
16    }
17    ll sum(int x, int y){
18        ll ans = 0ll;
19        for (int i = x; i > 0; i -= (i&(-i))){
20            for (int j = y; j > 0; j -= (j&(-j))){
21                ans += bit[i][j];
22            }
23        }
24        return ans;
25    }
26    ll query(int x1, int y1, int x2, int y2){
27        return sum(x2, y2) - sum(x2, y1 - 1) - sum(x1 - 1, y2) + sum(x1 - 1, y1 - 1);
28    }
29 };

```

5.6. Bit 2D Infinite Plane

```

1 struct BIT{
2     ll n;
3     vector<vector<ll>> xs;
4     vector<vector<ll>> bit;
5     BIT(ll nn){
6         n = nn;
7         xs.resize(n + 1);
8         bit.resize(n + 1);
9     }
10    void prepare(ll x, ll y){
11        for (ll i = x; i <= n; i += (i&(-i))){
12            xs[i].push_back(y);
13        }
14    }
15    void build(){
16        for (ll i = 1; i <= n; i++){
17            sort(xs[i].begin(), xs[i].end());
18            xs[i].erase(unique(xs[i].begin(), xs[i].end()), xs[i].end());
19            bit[i].assign(xs[i].size() + 1, 0);
20        }
21    }
22    void add(ll x, ll y, ll val){
23        for (ll i = x; i <= n; i += (i&(-i))){
24            ll p = lower_bound(xs[i].begin(), xs[i].end(), y) - xs[i].begin() + 1;
25            for (ll j = p; j < bit[i].size(); j += (j&(-j))){
26                bit[i][j] += val;
27            }
28        }
29    }
30    ll sum(ll x, ll y){
31        ll ans = 0ll;
32        for (ll i = x; i > 0; i -= (i&(-i))){
33            ll p = lower_bound(xs[i].begin(), xs[i].end(), y) - xs[i].begin();

```

```

34         for (ll j = p; j > 0; j -= (j&(-j))){
35             ans += bit[i][j];
36         }
37     }
38     return ans;
39 }
40 ll query(ll x1, ll y1, ll x2, ll y2){
41     return sum(x2-1,y2) - sum(x2-1, y1) - sum(x1 - 1, y2) + sum(x1 - 1, y1);
42 }
43
44 };
45 void solve(int tc){
46     int n,q; cin >> n >> q;
47     set<ll> X;
48     set<ll> Y;
49     vector<vector<ll>> points;
50     vector<vector<ll>> query;
51     repl(i,0,n){
52         ll x,y,w; cin >> x >> y >> w;
53         X.insert(x);
54         Y.insert(y);
55         points.push_back({x,y,w});
56     }
57     repl(i,0,q){
58         int t; cin >> t;
59         if(t == 0){
60             ll x,y,w; cin >> x >> y >> w;
61             X.insert(x);
62             Y.insert(y);
63             query.push_back({0,x,y,w});
64         }
65         else{
66             ll x1, y1, x2, y2;
67             cin >> x1 >> y1 >> x2 >> y2;
68             X.insert(x1);
69             X.insert(x2);
70             Y.insert(y1);
71             Y.insert(y2);
72             query.push_back({1,x1,y1,x2,y2});
73         }
74     }
75     map<ll,ll> equis, ye;
76     int i = 0;
77     for(ll x : X){
78         i++;
79         equis[x] = i;
80     }
81     i = 0;
82     for(ll x : Y){
83         i++;
84         ye[x] = i;
85     }
86     BIT bit((ll)X.size());
87     for(auto &x : points){
88         x[0] = equis[x[0]];
89         x[1] = ye[x[1]];
90         bit.prepare(x[0], x[1]);
91     }
92     for(auto &x : query){
93         if(x[0] == 0){
94             x[1] = equis[x[1]];
95             x[2] = ye[x[2]];
96             bit.prepare(x[1], x[2]);
97         }
98         else{
99             x[1] = equis[x[1]];
100             x[2] = ye[x[2]];
101             x[3] = equis[x[3]];
102             x[4] = ye[x[4]];
103         }
104     }
105     bit.build();
106     for(auto &x : points){
107         bit.add(x[0], x[1], x[2]);
108     }
109     for(auto &x : query){
110         if(x[0] == 0){
111             bit.add(x[1], x[2], x[3]);
112         }
113         else{
114             cout << bit.query(x[1],x[2],x[3],x[4]) << "\n";
115         }
116     }
117 }

```

5.7. Bit 3D

```

1  struct BIT {
2      int n;
3      int m;
4      int p;
5      vector < vector < vector < ll >>> bit;
6      BIT(int _n) {
7          n = _n + 10;
8          m = _n + 10;
9          p = _n + 10;
10         bit.assign(n, vector < vector < ll >> (n, vector < ll > (n, 0)));
11     }
12     void add(int x, int y, int z, int val) {
13         for (int i = x; i <= n; i += (i & (-i))) {
14             for (int j = y; j <= n; j += (j & (-j))) {
15                 for (int k = z; k <= n; k += (k & (-k))) {
16                     bit[i][j][k] += val;
17                 }
18             }
19         }
20     }
21     ll sum(int x, int y, int z) {
22         ll ans = 0;
23         for (int i = x; i > 0; i -= (i & (-i))) {
24             for (int j = y; j > 0; j -= (j & (-j))) {
25                 for (int k = z; k > 0; k -= (k & (-k))) {
26                     ans += bit[i][j][k];
27                 }
28             }
29         }
30         return ans;
31     }
32     ll query(int x1, int y1, int z1, int x2, int y2, int z2) {
33         return sum(x2, y2, z2) - sum(x1 - 1, y2, z2) - sum(x2, y2, z1 - 1) - sum(x2, y1 - 1, z2)
34             + sum(x1 - 1, y1 - 1, z2) + sum(x2, y1 - 1, z1 - 1) + sum(x1 - 1, y2, z1 - 1) -
35             sum(x1 - 1, y1 - 1, z1 - 1);
36     }
37 };

```

5.8. DSU Dynamic Representative

```

1  int n, q;
2  cin >> n >> q;
3  DSU dsu(n + q + 10);
4  vector<int> id(n + 1), box(n + q + 10);
5  int nodes = n;
6  iota(id.begin(), id.end(), 0);
7  iota(box.begin(), box.end(), 0);
8  for(int i = 0; i < q; i++){
9      int a, b;
10     cin >> a >> b;
11     int ori = id[a];
12     int ori2 = id[b];
13     dsu.parent[ori] = ori2;
14     nodes++;
15     id[a] = nodes;
16     box[nodes] = a;
17 }
18 for(int i = 1; i <= n; i++){
19     cout << box[dsu.find(i)] << " ";
20 }

```

5.9. Disjoint Set Union

```

1  struct DSU{
2      int num;
3      vector<int> parent;
4      vector<int> sizes;
5      DSU(){};
6      DSU(int n){
7          init(n);
8      }
9      void init(int n){
10         num = n;
11         parent.resize(n+1);
12         sizes.resize(n+1);
13         //si tienes index 0 inicializar desde 0
14         for(int i = 1; i <= n; i++){
15             parent[i] = i;

```



```

16         sizes[i] = 1;
17     }
18 }
19 int find(int a){
20     if(a == parent[a]) return a;
21     return parent[a] = find(parent[a]);
22 }
23 void join(int a, int b){
24     int x = find(a);
25     int y = find(b);
26     if(x == y) return;
27     num--;
28     if(sizes[x] < sizes[y]){
29         parent[x] = y;
30         sizes[y] += sizes[x];
31     }
32     else{
33         parent[y] = x;
34         sizes[x] += sizes[y];
35     }
36 }
37 }
38 };
39 //Sabroossus

```

5.10. Dynamic Connectivity Offline

```

1  struct Change {
2      int a, b, sz_a;
3  };
4
5  struct DSU {
6      vector<int> parent, sz;
7      vector<Change> history;
8      int components;
9
10     DSU(int n) {
11         parent.resize(n + 1);
12         sz.assign(n + 1, 1);
13         iota(parent.begin(), parent.end(), 0);
14         components = n;
15     }
16
17     int find(int x) {
18         while (x != parent[x])
19             x = parent[x];
20         return x;
21     }
22
23     bool unite(int a, int b) {
24         a = find(a);
25         b = find(b);
26         if (a == b) {
27             history.push_back({-1, -1, -1});
28             return false;
29         }
30         if (sz[a] < sz[b])
31             swap(a, b);
32         history.push_back({a, b, sz[a]});
33         parent[b] = a;
34         sz[a] += sz[b];
35         components--;
36         return true;
37     }
38
39     int snapshot() {
40         return history.size();
41     }
42
43     void rollback(int snap) {
44         while (history.size() > snap) {
45             auto [a, b, old_sz_a] = history.back();
46             history.pop_back();
47             if (a == -1)
48                 continue;
49             parent[b] = b;
50             sz[a] = old_sz_a;
51             components++;
52         }
53     }
54 };
55
56 struct Edge {
57     int u, v;

```

```

58 };
59
60 struct SegmentTree {
61     int n;
62     vector<vector<Edge>> tree;
63     SegmentTree(int n) {
64         this->n = n;
65         tree.resize(4 * n + 5);
66     }
67     void add(int node, int l, int r, int ql, int qr, Edge e) {
68         if (qr < l or r < ql)
69             return;
70         if (ql <= l and r <= qr) {
71             tree[node].push_back(e);
72             return;
73         }
74         int mid = (l + r) / 2;
75         add(node * 2, l, mid, ql, qr, e);
76         add(node * 2 + 1, mid + 1, r, ql, qr, e);
77     }
78     void add(int l, int r, Edge e) {
79         if (l > r)
80             return;
81         add(1, 1, n, l, r, e);
82     }
83 };
84
85 void dfs(int node, int l, int r, SegmentTree &seg, DSU &dsu, vector<int> &answer){
86     int snap = dsu.snapshot();
87     for (auto [u, v] : seg.tree[node])
88         dsu.unite(u, v);
89     if (l == r) {
90         answer[l] = dsu.components;
91     } else {
92         int mid = (l + r) / 2;
93         dfs(node * 2, l, mid, seg, dsu, answer);
94         dfs(node * 2 + 1, mid + 1, r, seg, dsu, answer);
95     }
96     dsu.rollback(snap);
97 }
98
99 void solve() {
100     int n, q;
101     cin >> n >> q;
102     if (q == 0)
103         return;
104     DSU dsu(n);
105     SegmentTree seg(q);
106     map<pair<int,int>, int> start;
107     vector<int> answer(q + 1);
108     vector<bool> query(q + 1);
109     for (int t = 1; t <= q; t++) {
110         char type;
111         cin >> type;
112         if (type == '?') {
113             query[t] = true;
114         } else {
115             int u, v;
116             cin >> u >> v;
117             if (u > v)
118                 swap(u, v);
119             if (type == '+') {
120                 start[{u, v}] = t;
121             } else {
122                 int L = start[{u, v}];
123                 seg.add(L, t - 1, {u, v});
124                 start.erase({u, v});
125             }
126         }
127     }
128     for (auto [edge, L] : start) {
129         auto [u, v] = edge;
130         seg.add(L, q, {u, v});
131     }
132     dfs(1, 1, q, seg, dsu, answer);
133     for (int t = 1; t <= q; t++) {
134         if (query[t])
135             cout << answer[t] << '\n';
136     }
137 }

```

5.11. Implicit Treap

```
1 // Mantiene las posiciones
```

```

2  struct Node {
3      int val , pri , sz ;
4      Node *l , *r;
5      Node ( int v ) {
6          val = v;
7          pri = rand () ;
8          sz = 1;
9          l = r = nullptr ;
10     }
11 };
12 using pnode = Node *;
13 int sz ( pnode t ) {
14     return t ? t -> sz : 0;
15 }
16 void upd (pnode t) {
17     if (t) t -> sz = 1 + sz (t -> l) + sz (t -> r);
18 }
19 void split ( pnode t , pnode &l , pnode &r , int k ) {
20     if (!t) {
21         l = r = nullptr ;
22         return ;
23     }
24     int leftSize = sz (t -> l);
25     if ( leftSize >= k ) {
26         split (t -> l , l , t -> l , k) ;
27         r = t;
28     }
29     else {
30         split (t -> r , t -> r , r , k - leftSize - 1) ;
31         l = t;
32     }
33     upd (t);
34 }
35 pnode merge ( pnode l , pnode r ) {
36     if (!l || !r ) return l ? l : r;
37
38     if (l -> pri > r -> pri ) {
39         l -> r = merge (l -> r , r);
40         upd (l );
41         return l;
42     }
43     else {
44         r -> l = merge (l , r -> l );
45         upd (r );
46         return r;
47     }
48 }
49 void insert ( pnode &root , int pos , int val ) {
50     pnode a , b;
51     split ( root , a , b , pos );
52     root = merge ( merge (a , new Node ( val )) , b) ;
53 }
54 void erase ( pnode &root , int pos ) {
55     pnode a , b , c;
56     split ( root , a , b , pos );
57     split (b , b , c , 1) ;
58     root = merge (a , c) ;
59 }
60 void print(pnode t) {
61     if (!t) return;
62
63     print(t->l);
64     cout << t->val << "□";
65     print(t->r);
66 }
67 int get(pnode t, int k) {
68     if (!t) return -1;
69
70     int leftSize = sz(t->l);
71
72     if (k < leftSize)
73         return get(t->l, k);
74
75     if (k == leftSize)
76         return t->val;
77
78     return get(t->r, k - leftSize - 1);
79 }

```

5.12. Ordered Set

```

1  #include <bits/stdc++.h>
2  #include <ext/pb_ds/assoc_container.hpp>
3  #include <ext/pb_ds/tree_policy.hpp>

```

```

4
5 using namespace std;
6 using namespace __gnu_pbds;
7
8 typedef tree<int, null_type, less<int>, rb_tree_tag, tree_order_statistics_node_update>
   ordered_set;
9
10 void solve() {
11     ordered_set st;
12
13     st.insert(10);
14     st.insert(30);
15     st.insert(20);
16
17     // 1. Elemento en la posicion K (0-indexed, de menor a mayor)
18     // find_by_order devuelve un iterador (puntero). Usamos '*' para el valor.
19     auto it = st.find_by_order(1);
20     if (it != st.end()) {
21         int elemento = *it; // Devuelve 20 de forma segura
22     }
23
24     // 2. Cuantos elementos son estrictamente menores que X
25     int menores_que_25 = st.order_of_key(25); // Devuelve 2 (el 10 y el 20)
26 }
27 // Sabrosssus

```

5.13. Range Bit

```

1 struct RangeBIT{
2     int n;
3     vector<ll> bit1, bit2;
4     RangeBIT(int nn){
5         n = nn;
6         bit1.assign(n + 1, 0);
7         bit2.assign(n + 1, 0);
8     }
9     void add_one(vector<ll> &b, int x, ll val){
10         for (int i = x; i <= n; i+= (i&(-i))){
11             b[i] += val;
12         }
13     }
14     void add(int l, int r, ll val){
15         add_one(bit1, l, val);
16         add_one(bit1, r + 1, -val);
17         add_one(bit2, l, l * val);
18         add_one(bit2, r + 1, -(r + 1) * val);
19     }
20     ll sum_one(const vector<ll> &b, int x){
21         ll ans = 0ll;
22         for (int i = x; i > 0; i -= (i&(-i))){
23             ans += b[i];
24         }
25         return ans;
26     }
27
28     ll sum(int x){
29         return (x + 1) * sum_one(bit1, x) - sum_one(bit2, x);
30     }
31     ll query(int l, int r){
32         return sum(r) - sum(l - 1);
33     }
34 };

```

5.14. Sparse Table

```

1 const int N = 1e5;
2 const int LOG = 20;
3 int v[N]; //values
4 int st[N][LOG]; //Sparse Table
5 int lg[N]; // log values
6 int n;
7 void build(){
8     for(int i = 0; i < n; i++) st[i][0] = v[i];
9
10    for(int i = 1; i < LOG; i++){
11        for(int j = 0; j < n - (1<<i) + 1; j++){
12            st[j][i] = min(st[j][i-1], st[j+(1<<(i-1))][i-1]);
13        }
14    }
15    lg[1] = 0;
16    for(int i = 2; i <= n; i++) lg[i] = lg[i/2] + 1;

```

```

17 }
18 int query(int L, int R){
19     int k = lg[R-L+1];
20     return min(st[L][k], st[R-(1<<k)+1][k]);
21 }
22 // v = [5, 2, 4, 7, 1, 3]
23 // query(1, 4)      [2, 4, 7, 1]

```

5.15. Treap

```

1 // Ordena los elementos por valor
2 struct Node {
3     int key, pri, sz;
4     Node *l, *r;
5     Node(int k) {
6         key = k;
7         pri = rand();
8         sz = 1;
9         l = r = nullptr;
10    }
11 };
12
13 using pnode = Node*;
14
15 int sz(pnode t) {
16     return t ? t->sz : 0;
17 }
18
19 void upd(pnode t) {
20     if (t) t->sz = 1 + sz(t->l) + sz(t->r);
21 }
22
23 void split(pnode t, pnode &l, pnode &r, int k) {
24     if (!t) {
25         l = r = nullptr;
26         return;
27     }
28
29     int leftSize = sz(t->l);
30
31     if (leftSize >= k) {
32         split(t->l, l, t->l, k);
33         r = t;
34     } else {
35         split(t->r, t->r, r, k - leftSize - 1);
36         l = t;
37     }
38
39     upd(t);
40 }
41
42 pnode merge(pnode l, pnode r) {
43     if (!l || !r) return l ? l : r;
44
45     if (l->pri > r->pri) {
46         l->r = merge(l->r, r);
47         upd(l);
48         return l;
49     } else {
50         r->l = merge(l, r->l);
51         upd(r);
52         return r;
53     }
54 }
55
56 void insert(pnode &root, int pos, int val) {
57     pnode a, b;
58     split(root, a, b, pos);
59     root = merge(merge(a, new Node(val)), b);
60 }
61
62 void erase(pnode &root, int pos) {
63     pnode a, b, c;
64     split(root, a, b, pos);
65     split(b, b, c, 1);
66     root = merge(a, c);
67 }
68
69 int get(pnode t, int k) {
70     if (!t) return -1;
71
72     int leftSize = sz(t->l);
73
74     if (k < leftSize)

```

```
75     return get(t->l, k);
76
77     if (k == leftSize)
78         return t->key;
79
80     return get(t->r, k - leftSize - 1);
81 }
82
83 void print(pnode t) {
84     if (!t) return;
85
86     print(t->l);
87     cout << t->key << " ";
88     print(t->r);
89 }
```

6. Dynamic Programing

6.1. Bounded Knapsack

```
1 // bounded_knapsack para DPs del estilo :
2 // dp[j - cw] + cv , w-> pesov-> valor y quieres maximizar el valor
3 int n, we;
4 cin >> n >> we;
5 vector<int> v, m, w, dp;
6 v.resize(n);
7 m.resize(n);
8 w.resize(n);
9 dp.assign(we+1, 0);
10 int sum = 0;
11 for(int i = 0; i < n; i++){
12     int a, b, c;
13     cin >> a >> b >> c;
14     v[i] = a;
15     w[i] = b;
16     m[i] = c;
17     vector<int> old = dp;
18     for(int r = 0; r < w[i] and r <= we; r++){
19         deque<pair<int, int>> dq;
20         for(int j = r; j <= we; j += w[i]){
21             int k = (j - r) / w[i];
22             while(!dq.empty() and dq.front().first < k - m[i]){
23                 dq.pop_front();
24             }
25             int cur = old[j] - k * v[i];
26             while(!dq.empty() && dq.back().second <= cur){
27                 dq.pop_back();
28             }
29             dq.push_back({k, cur});
30             dp[j] = dq.front().second + k * v[i];
31         }
32     }
33 }
34 cout << dp[we];
```

6.2. Kadane 2D

```
1 int n;
2 cin >> n;
3 vector < vector < int >> v(n);
4 repl(i, 0, n) {
5     repl(j, 0, n) {
6         int x;
7         cin >> x;
8         v[i].push_back(x);
9     }
10 }
11 int maxi = INT_MIN;
12 repl(ini, 0, n) {
13     vector < int > pref(n, 0);
14     repl(i, ini, n) {
15         repl(j, 0, n) {
16             pref[j] += v[i][j];
17         }
18         int cur = 0;
19         repl(j, 0, n) {
20             cur = max(pref[j], pref[j] + cur);
21             maxi = max(maxi, cur);
22         }
23     }
24 }
```

```

22     }
23 }
24 }

```

6.3. Kadane 3D

```

1  int n;
2  cin >> n;
3
4  repl(i, 0, n) {
5      repl(j, 0, n) {
6          repl(k, 0, n) {
7              cin >> v[i][j][k];
8          }
9      }
10 }
11 int maxi = INT_MIN;
12 repl(k_ini, 0, n) {
13     vector < vector < int >> pref2(n, vector < int > (n, 0));
14     repl(k, k_ini, n) {
15         repl(i, 0, n) {
16             repl(j, 0, n) {
17                 pref2[i][j] += v[i][j][k];
18             }
19         }
20         repl(ini, 0, n) {
21             vector < int > pref(n, 0);
22             repl(i, ini, n) {
23                 repl(j, 0, n) {
24                     pref[j] += pref2[i][j];
25                 }
26                 int cur = 0;
27                 repl(u, 0, n) {
28                     cur = max(pref[u], cur + pref[u]);
29                     maxi = max(maxi, cur);
30                 }
31             }
32         }
33     }
34 }
35 }
36
37 cout << maxi << "\n";

```

6.4. SOS DP

```

1  void sos_dp(vector<ll>& dp, int K){
2      // k es el tamaño del arreglo :v
3      for(int bit = 0; bit < K; bit++){
4          for(int mask = 0; mask < (1 << K); mask++){
5              if(mask & (1 << bit)){
6                  dp[mask] += dp[mask ^ (1 << bit)];
7              }
8              // si quieres que sea un sos dp de maximos
9              // dp[mask] = max(dp[mask], dp[mask ^ (1 << bit)]);
10         }
11     }
12 }

```

6.5. Fastfibo

```

1  #include <bits/stdc++.h>
2  #define ll long long
3
4  using namespace std;
5
6  const ll MOD = 1e9 + 7;
7
8  struct Matrix {
9      ll a[2][2];
10     Matrix() {
11         memset(a, 0, sizeof(a));
12     }
13 };
14
15 Matrix multiply(Matrix A, Matrix B) {
16     Matrix C;

```

```

17     for (int i = 0; i < 2; i++) {
18         for (int j = 0; j < 2; j++) {
19             for (int k = 0; k < 2; k++) {
20                 C.a[i][j] = (C.a[i][j] + A.a[i][k] * B.a[k][j]) % MOD;
21             }
22         }
23     }
24     return C;
25 }
26
27 Matrix power(Matrix A, ll n) {
28     Matrix R;
29     R.a[0][0] = 1;
30     R.a[1][1] = 1;
31     while (n) {
32         if (n & 1)
33             R = multiply(R, A);
34         A = multiply(A, A);
35         n >>= 1;
36     }
37     return R;
38 }
39
40 ll fibonacci(ll n) {
41     if (n == 0) return 0;
42     Matrix A;
43     A.a[0][0] = 1;
44     A.a[0][1] = 1;
45     A.a[1][0] = 1;
46     A.a[1][1] = 0;
47     Matrix R = power(A, n - 1);
48     return R.a[0][0];
49 }
50
51 int main() {
52     ll n;
53     cin >> n;
54     cout << fibonacci(n) << '\n';
55 }

```

7. Flow

7.1. Dinic

```

1  struct Dinic
2  {
3      const int INF = 1e9;
4
5      struct flowEdge
6      {
7          int to, rev, f, cap;
8      };
9
10     vector<vector<flowEdge>> G;
11     vector<int> work, lvl, Q;
12
13     int S, T, nodes;
14
15     Dinic(){}
16     Dinic(int n, int s, int t){
17         S = s;
18         T = t;
19         nodes = n;
20         G.clear();
21         G.resize(n);
22         Q.resize(n);
23     }
24     void addEdge(int st, int en, int cap){
25         flowEdge A = {en, (int)G[en].size(), 0, cap};
26         flowEdge B = {st, (int)G[st].size(), 0, 0};
27         G[st].push_back(A);
28         G[en].push_back(B);
29     }
30     int dfs(int v, int f){
31         if(v == T or f == 0) return f;
32         for(int &i = work[v]; i < G[v].size(); i++){
33             flowEdge &e = G[v][i];
34             int u = e.to;
35             if(e.cap <= e.f or lvl[u] != lvl[v] + 1) continue;
36             int df = dfs(u, min(f, e.cap - e.f));
37             if(df){

```



```

38         e.f += df;
39         G[u][e.rev].f -= df;
40         return df;
41     }
42 }
43     return 0;
44 }
45 bool bfs(){
46     int qt = 0;
47     Q[qt++] = S;
48     lvl.assign(nodes, -1);
49     lvl[S] = 0;
50     for(int i = 0; i < qt; i++){
51         int u = Q[i];
52         for(flowEdge &e : G[u]){
53             int v = e.to;
54             if(e.cap <= e.f or lvl[v] != -1) continue;
55             lvl[v] = lvl[u] + 1;
56             Q[qt++] = v;
57         }
58     }
59     return lvl[T] != -1;
60 }
61 int maxFlow(){
62     int flow = 0;
63     while(bfs()){
64         work.assign(nodes, 0);
65         while(true){
66             int df = dfs(S, INF);
67             if(df == 0) break;
68             flow += df;
69         }
70     }
71     return flow;
72 }
73
74 };

```

8. Graph Algorithms

8.1. Dijkstra

```

1  const int N = 1e5;
2
3  int vis[N];
4  vector<pair<int,int>> G[N];
5
6  void dijkstra(int x){
7      priority_queue<pair<int,int>, vector<pair<int,int>>, greater<pair<int,int>>> pq;
8      pq.push({0,x});
9      vis[x] = 0;
10     while(!pq.empty()){
11         auto y = pq.top();
12         pq.pop();
13         int u = y.second;
14         int cost = y.first;
15         for(auto v : G[u]){
16             int cur = cost + v.second;
17             if(cur < vis[v.first]){
18                 vis[v.first] = cur;
19                 pq.push({cur,v.first});
20             }
21         }
22     }
23 }

```

8.2. Kruskal MST

```

1  const int N = 1e5;
2  struct DSU{
3      int num;
4      vector<int> parent;
5      vector<int> sizes;
6      DSU(){};
7      DSU(int n){
8          init(n);
9      }
10     void init(int n){

```

```

11     num = n;
12     parent.resize(n+1);
13     sizes.resize(n+1);
14     for(int i = 1; i <= n ; i++){
15         parent[i] = i;
16         sizes[i] = 1;
17     }
18 }
19 int find(int a){
20     if(a == parent[a]) return a;
21     return parent[a] = find(parent[a]);
22 }
23 void join(int a, int b){
24     int x = find(a);
25     int y = find(b);
26     if(x == y) return;
27     num--;
28     if(sizes[x] < sizes[y]){
29         parent[x] = y;
30         sizes[y] += sizes[x];
31     }
32     else{
33         parent[y] = x;
34         sizes[x] += sizes[y];
35     }
36 }
37 };
38 int n;
39 vector<pair<int,pair<int,int>>> List;
40 int Kruskal_MST(){
41     sort(List.begin(),List.end());
42     DSU dsu(n);
43     int mst = 0, t = 1;
44     for(int i = 0; i < List.size(); i++){
45         int w = List[i].first; //weight
46         int a = List[i].second.first; //at
47         int b = List[i].second.second; //to
48         if(dsu.find(a) != dsu.find(b)){
49             dsu.join(a,b);
50             mst += w;
51             t++;
52         }
53         if(t == n) break;
54     }
55     return mst;
56 }

```

8.3. Topological Sort Kahn Algorithm

```

1  const int N = 1e5;
2
3  int indegree[N];
4  vector<int> G[N];
5  int n;
6  vector<int> Kahn(){
7      queue<int> q;
8      vector<int> ans;
9      for(int i = 0; i < n; i++){
10         if(indegree[i] == 0) q.push(i);
11     }
12     while (!q.empty()){
13         int v = q.front();
14         q.pop();
15         ans.push_back(v);
16         for(int u : G[v]){
17             indegree[u]--;
18             if(indegree[u] == 0){
19                 q.push(u);
20             }
21         }
22     }
23     //if(ans.size() != n) grafo no DAG
24     return ans;
25 }

```

8.4. Kosaraju

```

1  struct Kosaraju {
2      int n, components;
3      vector<vector<int>> graph, invgraph;
4      vector<int> vis, order, component;

```

```

5
6     Kosaraju(int n) : n(n) {
7         graph.resize(n + 1);
8         invgraph.resize(n + 1);
9         vis.resize(n + 1);
10        component.resize(n + 1);
11    }
12
13    void add_edge(int u, int v) {
14        graph[u].push_back(v);
15        invgraph[v].push_back(u);
16    }
17
18    void dfs1(int u) {
19        vis[u] = 1;
20        for (int v : graph[u]) {
21            if (!vis[v])
22                dfs1(v);
23        }
24        order.push_back(u);
25    }
26
27    void dfs2(int u, int id) {
28        vis[u] = 1;
29        component[u] = id;
30        for (int v : invgraph[u]) {
31            if (!vis[v])
32                dfs2(v, id);
33        }
34    }
35    void build() {
36        for (int u = 1; u <= n; u++) {
37            if (!vis[u])
38                dfs1(u);
39        }
40        reverse(order.begin(), order.end());
41        fill(vis.begin(), vis.end(), 0);
42        components = 0;
43        for (int u : order) {
44            if (!vis[u]) {
45                components++;
46                dfs2(u, components);
47            }
48        }
49    }
50 };

```

8.5. Kuhn

```

1  int n;
2  vector<vector<int>> G;
3  vector<int> mt;
4  vector<bool> used;
5  bool try_kuhn(int v){
6      if(used[v]) return false;
7      used[v] = true;
8      for(int to : G[v]){
9          if(mt[to] == -1 or try_kuhn(mt[to])){
10             mt[to] = v;
11             return true;
12         }
13     }
14     return false;
15 }
16 void solve(int tc){
17     mt.assign(n, -1);
18     repl(v,0,n){
19         used.assign(n, false);
20         try_kuhn(v);
21     }
22 }

```

8.6. 2SAT

```

1  struct TwoSAT {
2      int n, nodes;
3      vector<vector<int>> graph, invgraph;
4      vector<int> vis, order, component;
5      vector<bool> ans;
6
7      TwoSAT(int n) : n(n), nodes(2 * n) {

```

```

8      graph.resize(nodes);
9      invgraph.resize(nodes);
10     vis.resize(nodes);
11     component.resize(nodes);
12     ans.resize(n);
13 }
14
15 int id(int x, bool value) {
16     return 2 * x + value;
17 }
18
19 int neg(int x) {
20     return x ^ 1;
21 }
22
23 void add_implication(int a, int b) {
24     graph[a].push_back(b);
25     invgraph[b].push_back(a);
26 }
27
28 void add_or(int a, int b) {
29     add_implication(neg(a), b);
30     add_implication(neg(b), a);
31 }
32
33 void add_xor(int a, int b) {
34     add_or(a, b);
35     add_or(neg(a), neg(b));
36 }
37
38 void add_equivalence(int a, int b) {
39     add_implication(a, b);
40     add_implication(b, a);
41     add_implication(neg(a), neg(b));
42     add_implication(neg(b), neg(a));
43 }
44
45 void dfs1(int u) {
46     vis[u] = 1;
47     for (int v : graph[u]) {
48         if (!vis[v])
49             dfs1(v);
50     }
51     order.push_back(u);
52 }
53
54 void dfs2(int u, int id) {
55     vis[u] = 1;
56     component[u] = id;
57     for (int v : invgraph[u]) {
58         if (!vis[v])
59             dfs2(v, id);
60     }
61 }
62
63 bool solve() {
64     for (int u = 0; u < nodes; u++) {
65         if (!vis[u])
66             dfs1(u);
67     }
68     reverse(order.begin(), order.end());
69     fill(vis.begin(), vis.end(), 0);
70     int cnt = 0;
71     for (int u : order) {
72         if (!vis[u]) {
73             dfs2(u, cnt++);
74         }
75     }
76     for (int i = 0; i < n; i++) {
77         if (component[2 * i] == component[2 * i + 1])
78             return false;
79         ans[i] = component[2 * i] > component[2 * i + 1];
80     }
81     return true;
82 }
83 };

```

9. Tree Algorithms

9.1. Euler Tour

```

1  const int N = 1e5 + 10;
2  vector<int> path;
3  vector<int> G[N];
4  void EulerTour(int node, int parent){
5      path.pb(node);
6      for(int u : G[node]){
7          if(u != parent){
8              EulerTour(u,node);
9          }
10     }
11     path.pb(node);
12 }
13
14 const int N = 1e5 + 10;
15 int tin[N];
16 int tout[N];
17 int flat[N];
18 int timer = 1;
19 void EulerTour(int node, int parent){
20     tin[node] = timer;
21     flat[timer] = node;
22     timer++;
23     for(int u : G[node]){
24         if(u != parent){
25             EulerTour(u, node);
26         }
27     }
28     tout[node] = timer;
29 }

```

9.2. Binary Lifting

```

1  const int N = 1e5;
2  const int LOG = 20;
3  int up[N][LOG];
4  vector<int> G[N];
5  //Find parents
6  void dfs(int u){
7      for(int v : G[u]){
8          up[v][0] = u;
9          for(int j = 1; j < LOG; j++){
10             up[v][j] = up[up[v][j-1]][j-1];
11         }
12         dfs(v);
13     }
14 }
15 int get_K_parent(int a, int k){
16     for(int j = LOG-1; j >= 0; j++){
17         if(k & (1<<j)){
18             a = up[a][j];
19         }
20     }
21     return a;
22 }

```

9.3. Centroid Decomposition

```

1  //EJERCICIO IMPLEMENTACION XENIA AND TREE
2  //Nodo rojo mas cercano a un nodo v
3
4  const int N = 2e5+10;
5  vector<int> G[N];
6  int tag[N]; // time of discovery of centroidi
7  int fat[N]; // father in centroid decomposition
8  int szt[N]; // size of subtree
9
10 int best[N];
11 int niv[N];
12 int dist[30][N]; //dist[nivel][nodo];
13
14 void dfs_level(int node, int p, int niv, int prof){
15     dist[niv][node] = prof;
16     for(auto y : G[node]){
17         if(y != p and tag[y] < 0){
18             dfs_level(y, node, niv, prof+1);
19         }
20     }
21 }
22 // -----CENTROID-----
23 int calcsz(int node, int p){
24     szt[node] = 1;

```

```

25     for(auto y : G[node]){
26         if(y != p and tag[y] < 0){
27             szt[node] += calcsz(y, node);
28         }
29     }
30     return szt[node];
31 }
32 int ccnt = 0;
33 void cdfs(int node, int p, int nivel = 0, int sz = -1){
34     if(sz < 0) sz = calcsz(node, -1);
35     for(auto y : G[node]){
36         if(tag[y] < 0 and szt[y] * 2 >= sz){
37             szt[node] = 0;
38             cdfs(y, p, nivel, sz);
39             return;
40         }
41     }
42     tag[node] = ccnt++;
43     fat[node] = p;
44     niv[node] = nivel;
45     dfs_level(node, -1, nivel, 0);
46     for(auto y : G[node]){
47         if(tag[y] < 0){
48             cdfs(y, node, nivel+1);
49         }
50     }
51 }
52
53 int n;
54 void centroid(){
55     ccnt = 0;
56     repl(i,1,n+1){
57         tag[i] = -1;
58     }
59     cdfs(1, -1);
60 }
61
62 //-----
63
64 void update(int node, int rojo){
65     int nivel = niv[node];
66     int cost = dist[nivel][rojo];
67     best[node] = min(best[node], cost);
68     int p = fat[node];
69     if(p > 0){
70         update(p, rojo);
71     }
72 }
73
74 int improve(int node, int cur){
75     if(node <= 0){
76         return 1e9;
77     }
78     int nivel = niv[node];
79     int cost = best[node] + dist[nivel][cur];
80     int p = fat[node];
81     return min(cost, improve(p, cur));
82 }
83
84 void solve(){
85     int qq;
86     cin >> n >> qq;
87     repl(i,0,30){
88         repl(j,1,n+1){
89             dist[i][j] = -1;
90             best[j] = 1e9;
91         }
92     }
93     repl(i,0,n-1){
94         int a, b;
95         cin >> a >> b;
96         G[a].push_back(b);
97         G[b].push_back(a);
98     }
99     centroid();
100    update(1, 1);
101    while(qq--){
102        int q, node;
103        cin >> q >> node;
104        if(q == 1){
105            update(node, node);
106        }
107        else{
108            int res = best[node];
109            int res2 = improve(node, node);
110            cout << min(res, res2) << '\n';

```

```

111     }
112   }
113 }
114
115 int main(){
116     velocito;
117     int t = 1;
118     // cin >> t;
119     while(t--){
120         solve();
121     }
122     return 0;
123 }
124 //Sabrossus
125 //TC Argentina

```

9.4. HLD

```

1  struct HLD {
2      int n, act;
3      vector<vector<int>> adj;
4      vector<int> parent, depth, heavy, head, pos;
5      vector<int> val;
6      vector<int> seg;
7      int tam;
8      HLD(int n) : n(n) {
9          adj.resize(n+1);
10         parent.resize(n+1);
11         depth.resize(n+1);
12         heavy.assign(n+1, -1);
13         head.resize(n+1);
14         pos.resize(n+1);
15         val.resize(n+1);
16         act = 0;
17     }
18
19     void add_edge(int u, int v) {
20         adj[u].push_back(v);
21         adj[v].push_back(u);
22     }
23
24     int dfs(int u, int p) {
25         parent[u]=p;
26         int size=1, max_sub=0;
27         for (int v : adj[u]){
28             if(v==p)continue;
29             depth[v]=depth[u]+1;
30             int sub=dfs(v,u);
31             if(sub>max_sub){
32                 max_sub=sub;
33                 heavy[u]=v;
34             }
35             size+=sub;
36         }
37         return size;
38     }
39
40     void decompose(int u, int h) {
41         head[u]=h;
42         pos[u]=act++;
43         if(heavy[u]!=-1)decompose(heavy[u], h);
44         for(int v : adj[u]){
45             if(v!=parent[u] and v!=heavy[u])decompose(v,v);
46         }
47     }
48
49     void build(){
50         tam=1;
51         while (tam < n) tam<=1;
52         seg.assign(2*tam,inf);
53         for (int i = 1; i <= n; i++){
54             seg[tam+pos[i]]=val[i];
55         }
56         for (int i = tam - 1; i > 0; i--){
57             seg[i]=max(seg[2*i],seg[2*i+1]);
58         }
59     }
60
61     void update(int u, int value) {
62         int i=tam+pos[u];
63         seg[i]=value;
64         for(i>>=1; i; i>>=1){
65             seg[i]=max(seg[2*i],seg[2*i+1]);
66         }

```

```

67     }
68
69     int query(int l, int r) {
70         int res=inf;
71         l+=tam;
72         r+=tam;
73         while(l <= r){
74             if(l&1)res=max(res,seg[l++]);
75             if(!(r&1))res=max(res,seg[r--]);
76             l>>=1;
77             r>>=1;
78         }
79         return res;
80     }
81
82     void init(){
83         int root=1;
84         dfs(root, 0);
85         decompose(root,root);
86         build();
87     }
88
89     int query_path(int a, int b){
90         int res=inf;
91         while(head[a]!=head[b]){
92             if(depth[head[a]]<depth[head[b]])swap(a,b);
93             res=max(res,query(pos[head[a]],pos[a]));
94             a=parent[head[a]];
95         }
96         if(depth[a]>depth[b])swap(a,b);
97         res=max(res,query(pos[a],pos[b]));
98         return res;
99     }
100 };
101
102 void solve(){
103     int n,q;
104     cin>>n>>q;
105     HLD hld(n);
106     for(int i = 1 ; i <= n ; i++){
107         cin>>hld.val[i];
108     }
109     for(int i = 0; i < n-1 ; i++){
110         int a,b;
111         cin>>a>>b;
112         hld.add_edge(a,b);
113     }
114     hld.init();
115     while(q--){
116         int t;
117         cin>>t;
118         if(t==1){
119             int s,x;
120             cin>>s>>x;
121             hld.update(s,x);
122         }
123         else{
124             int a,b;
125             cin>>a>>b;
126             cout<<hld.query_path(a,b)<<' ';<
127         }
128     }
129 }

```

9.5. Lowest Common Ancestor

```

1  const int N = 1e5;
2  const int LOG = 20;
3  vector<int> G[N];
4  int depth[N];
5  int up[N][LOG];
6  // Assign levels and get parents
7  void dfs(int u){
8      for(int v : G[u]){
9          depth[v] = depth[u] + 1;
10         up[v][0] = u;
11         for(int j = 1; j < LOG; j++){
12             up[v][j] = up[up[v][j-1]][j-1];
13         }
14         dfs(v);
15     }
16 }
17 int get_lca(int a, int b){
18     if(depth[a] < depth[b]) swap(a,b);

```



```

19     int k = depth[a] - depth[b];
20     for(int j = LOG-1; j >= 0; j--){
21         if(k & (1<<j)){
22             a = up[a][j];
23         }
24     }
25     if(a == b) return a;
26     for(int j = LOG-1; j >= 0; j--){
27         if(up[a][j] != up[b][j]){
28             a = up[a][j];
29             b = up[b][j];
30         }
31     }
32     return up[a][0];
33 }

```

10. Utilities

10.1. Prime Numbers

```

1      2      3      5      7      11      13      17      19      23      29
2      31      37      41      43      47      53      59      61      67      71
3      73      79      83      89      97      101      103      107      109      113
4     127     131     137     139     149     151     157     163     167     173
5     179     181     191     193     197     199     211     223     227     229
6     233     239     241     251     257     263     269     271     277     281
7     283     293     307     311     313     317     331     337     347     349
8     353     359     367     373     379     383     389     397     401     409
9     419     421     431     433     439     443     449     457     461     463
10    467     479     487     491     499     503     509     521     523     541
11    547     557     563     569     571     577     587     593     599     601
12    607     613     617     619     631     641     643     647     653     659
13    661     673     677     683     691     701     709     719     727     733
14    739     743     751     757     761     769     773     787     797     809
15    811     821     823     827     829     839     853     857     859     863
16    877     881     883     887     907     911     919     929     937     941
17    947     953     967     971     977     983     991     997     1009     1013
18   1019    1021    1031    1033    1039    1049    1051    1061    1063    1069
19   1087    1091    1093    1097    1103    1109    1117    1123    1129    1151
20   1153    1163    1171    1181    1187    1193    1201    1213    1217    1223
21   1229    1231    1237    1249    1259    1277    1279    1283    1289    1291
22   1297    1301    1303    1307    1319    1321    1327    1361    1367    1373
23   1381    1399    1409    1423    1427    1429    1433    1439    1447    1451
24   1453    1459    1471    1481    1483    1487    1489    1493    1499    1511
25   1523    1531    1543    1549    1553    1559    1567    1571    1579    1583
26   1597    1601    1607    1609    1613    1619    1621    1627    1637    1657
27   1663    1667    1669    1693    1697    1699    1709    1721    1723    1733
28   1741    1747    1753    1759    1777    1783    1787    1789    1801    1811
29   1823    1831    1847    1861    1867    1871    1873    1877    1879    1889
30   1901    1907    1913    1931    1933    1949    1951    1973    1979    1987
31
32          970'997'997971'483      921'281'269      999'279'733
33 1'000'000'0091'000'000'021 1'000'000'4091'005'012'527
34 10005512850571600911

```

10.2. ASCII TABLE

```

1  ASCII QUICK REFERENCE
2  =====
3
4  CHARACTER RANGES
5  -----
6  '0' ... '9'      48 ... 57      -> c - '0'
7  'A' ... 'Z'      65 ... 90      -> c - 'A'
8  'a' ... 'z'      97 ... 122     -> c - 'a'
9
10 RELATIONSHIPS
11 -----
12 'a' - 'A' = 32
13 'A' + 32 = 'a'
14 'a' - 32 = 'A'
15
16 DIGITS
17 -----
18 '0' = 48      '5' = 53
19 '1' = 49      '6' = 54
20 '2' = 50      '7' = 55
21 '3' = 51      '8' = 56

```

```

22 '4' = 52      '9' = 57
23
24 UPPERCASE
25 -----
26 A 65  B 66  C 67  D 68  E 69  F 70  G 71
27 H 72  I 73  J 74  K 75  L 76  M 77  N 78
28 O 79  P 80  Q 81  R 82  S 83  T 84  U 85
29 V 86  W 87  X 88  Y 89  Z 90
30
31 LOWERCASE
32 -----
33 a 97  b 98  c 99  d 100 e 101 f 102 g 103
34 h 104 i 105 j 106 k 107 l 108 m 109 n 110
35 o 111 p 112 q 113 r 114 s 115 t 116 u 117
36 v 118 w 119 x 120 y 121 z 122
37
38 COMMON SYMBOLS
39 -----
40 SPACE 32      ! 33      " 34  _ 35  $ 36
41 % 37  & 38  ' 39  ( 40  ) 41
42 * 42  + 43  , 44  - 45  . 46
43 / 47  : 58  ; 59  < 60  = 61
44 > 62  ? 63  @ 64  [ 91  \ 92
45 ] 93  ^ 94  _ 95  ` 96  { 123
46 | 124 } 125 ~ 126
47
48 USEFUL TRICKS
49 -----
50 int ux = uc - '0'; // digit -> integer
51 char c = '0' + ux; // integer -> digit
52
53 int lx = lc - 'a'; // lowercase -> [0,25]
54 int ux = uc - 'A'; // uppercase -> [0,25]
55
56 islower(c)
57 isupper(c)
58 isdigit(c)
59 isalpha(c)
60 isalnum(c)
61 isspace(c)
62 tolower(c)
63 toupper(c)
64 =====

```

10.3. Int128

```

1  // --- LECTURA B SICA ---
2  __int128 read128() {
3      string s;
4      cin >> s;
5      __int128 n = 0;
6      int sign = 1;
7      size_t start = 0;
8
9      if (!s.empty() && s[0] == '-') {
10         sign = -1;
11         start = 1;
12     }
13
14     for (size_t i = start; i < s.length(); ++i) {
15         n = n * 10 + (s[i] - '0');
16     }
17
18     return n * sign;
19 }
20
21 // --- IMPRESI N B SICA ---
22 void print128(__int128 n) {
23     if (n == 0) {
24         cout << 0;
25         return;
26     }
27     if (n < 0) {
28         cout << '-';
29         n = -n;
30     }
31     string s;
32     while (n > 0) {
33         s += (char)('0' + (n % 10));
34         n /= 10;
35     }
36     reverse(s.begin(), s.end());
37     cout << s;
38 }

```

10.4. Merge Sort

```

1 // Rango de uso: 0-indexed.
2 // Llamar en el solve como: mergeSort(v, 0, n - 1);
3 ll ans = 0;
4
5 void mergeSort(vector<ll> &v, int low, int high) {
6     if (low == high) return;
7
8     int mid = (low + high) >> 1;
9     mergeSort(v, low, mid);
10    mergeSort(v, mid + 1, high);
11
12    queue<ll> L, H;
13    for (int i = low; i < mid + 1; i++) {
14        L.push(v[i]);
15    }
16    for (int i = mid + 1; i < high + 1; i++) {
17        H.push(v[i]);
18    }
19
20    for (int i = low; i < high + 1; i++) {
21        if (L.size() == 0) {
22            v[i] = H.front();
23            H.pop();
24        }
25        else if (H.size() == 0) {
26            v[i] = L.front();
27            L.pop();
28        }
29        else {
30            if (L.front() <= H.front()) {
31                v[i] = L.front();
32                L.pop();
33            }
34            else {
35                v[i] = H.front();
36                H.pop();
37                ans += L.size(); // Inversion detectada
38            }
39        }
40    }
41 }

```

10.5. Prefix Sum 2D

```

1 //1 indexed
2 const int N = 1005;
3 int v[N][N];
4 int acu[N][N];
5 int n, m;
6
7 void solve() {
8     //limpiar la acu para cada TC
9
10    // Construcción: acumula 1 si el elemento es exactamente 1
11    for (int i = 1; i <= n; i++) {
12        for (int j = 1; j <= m; j++) {
13            int val = (v[i][j] == 1 ? 1 : 0);
14            acu[i][j] = val + acu[i - 1][j] + acu[i][j - 1] - acu[i - 1][j - 1];
15        }
16    }
17
18    // Esquinas de consulta:
19    // (a, b) -> Superior Izquierda
20    // (c, d) -> Inferior Derecha
21    int a = 2, b = 2, c = 4, d = 4; // Valores de ejemplo
22
23    int ans = acu[c][d] - acu[a - 1][d] - acu[c][b - 1] + acu[a - 1][b - 1];
24    cout << ans << '\n';
25 }

```

10.6. Priority Queue Min Heap

```

1 priority_queue<int, vector<int>, greater<int>> pq;

```

10.7. Random

```

1 mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());
2 // mt19937_64 para 64 bits usar uint64_t o unsigned long long
3 cout << rng() << endl;
4 shuffle(v.begin(),v.end(),rng);

```

10.8. Plantilla Python

```

1 import sys
2 from bisect import bisect_left, bisect_right, insort
3 from collections import defaultdict, Counter, deque
4 from heapq import heappush, heappop, heapify
5 from math import gcd, lcm, isqrt
6
7 input = sys.stdin.readline
8
9
10 def solve():
11     n = int(input()) #leer numero
12     a,b,c= map(int, input().split()) #leer numeros
13     s = input().strip() #leer cadenas
14     v = list(map(int, input().split())) #leer lista de enteros
15
16
17 t = 1
18 # t = int(input())
19 for _ in range(t):
20     solve()
21
22 # C++                                Python
23
24 # vector                                list
25 # array                                list
26 # string                               str
27
28 # set                                  set
29 # map                                  dict
30 # multiset                             Counter
31 # unordered_map                        dict
32 # unordered_set                       set
33
34 # priority_queue                       heapq
35 # queue                                deque
36 # deque                                deque
37
38 # lower_bound                          bisect_left
39 # upper_bound                          bisect_right
40 # sort                                  list.sort()
41 # binary_search                        bisect_left/right
42
43 # pair                                  tuple
44 # struct                               class / dataclass
45
46 # gcd                                  math.gcd
47 # lcm                                  math.lcm
48
49 # push_back                            append
50 # push_front                          appendleft
51 # pop_back                             pop
52 # pop_front                           popleft
53
54 # size()                              len()
55 # empty()                             not x

```

10.9. Pragma

```

1 #pragma GCC optimize("O3,unroll-loops")
2 #pragma GCC target("avx2,bmi,bmi2,lzcnt,popcnt")

```